

Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Wednesday, April 10th, 2024, 7:30pm-9:00pm

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

Dr. Péter Juhász	Dr. Sean Paul	Dr. Terry Harris
Awildo Gutierrez, M 7:45 AM	Jiaming Xu, T 11:00 AM	Ryan Tamura, T 12:05 PM
Awildo Gutierrez, M 8:50 AM	Jiaming Xu, T 12:05 PM	Rune Chen, T 1:20 PM
Hongyu Zhu, W 9:55 AM	Jiaming Xu, R 11:00 AM	Ryan Tamura, R 12:05 PM
Awildo Gutierrez, W 7:45 AM	Zaidan Wu, R 12:05 PM	Hanzhang Mao, R 1:20 PM
Hongyu Zhu, W 8:50 AM	Zaidan Wu, R 1:20 PM	Connor Simpson, T 2:25 PM
Jacob Denson, M 9:55 AM		Ryan Tamura, T 11:00 AM
Carsen Grote, M 8:50 AM		Lauren Neudorf, R 2:25 PM
Carsen Grote, M 9:55 AM		Connor Simpson, T 12:05 PM
Jacob Denson, W 8:50 AM		Hanzhang Mao, R 12:05 PM
Carsen Grote, W 9:55 AM		Connor Simpson, T 9:55 AM
Jacob Denson, W 9:55 AM		

Name: Pepe Silvia Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of eight problems for a total of 50 points; the exam is 6 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (in x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (5 points, 1 point each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) Any three nonzero vectors in $\mathbb{R}^2$ will span $\mathbb{R}^2$.

- ☐ True
☒ False

b.) If A is a 3×6 matrix, then the dimension of the nullspace of A is at least 3.

- ☒ True
☐ False

c.) Let V be a vector space, and let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span V . Let $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ also span V . Then $n = m$.

- ☐ True
☒ False

d.) Any set of 4 polynomials in P_2 will be linearly dependent.

- ☒ True
☐ False

e.) If $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a one-to-one (also called injective) linear transformation, then $T(\mathbf{x}) = \mathbf{b}$ has a solution for any $\mathbf{b}$ in $\mathbb{R}^3$.

- ☒ True
☐ False

2. (9 points total) (This problem has overflow room on the next page) Consider $A = \begin{bmatrix} -7 & 4 \\ -12 & 7 \end{bmatrix}$.
You can use the fact that A has eigenvalues 1 and -1 .

a. (6 points) Diagonalize A (in particular, find D , P , and P^{-1}).

$$\underline{\lambda=1}: \begin{bmatrix} 8 & 4 \\ 12 & -6 \end{bmatrix} \xrightarrow{2a-b=0 \Rightarrow 2a=b} \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ e-vector.}$$

$$\underline{\lambda=-1}: \begin{bmatrix} 6 & -4 \\ 12 & -8 \end{bmatrix} \xrightarrow{3a-2b=0 \Rightarrow 3a=2b} \begin{bmatrix} 3 & -2 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightsquigarrow P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \quad \det P = 3-4 = -1.$$

$$P^{-1} = -1 \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & -b \\ -c & a \end{bmatrix} = \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$AP = PD \Rightarrow P^{-1}AP = D, \quad A = PDP^{-1}$$

- b. (3 points) Use your answer to part (a) to compute $A^{73}\mathbf{v}$, where $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

$$\begin{aligned} A^{73} &= (PDP^{-1})^{73} = PD^{73}P^{-1} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}^{73} \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -7 & 4 \\ -12 & 7 \end{bmatrix} = A \end{aligned}$$

$$\begin{bmatrix} -7 & 4 \\ -12 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -14+4 \\ -24+7 \end{bmatrix} = \begin{bmatrix} -10 \\ -17 \end{bmatrix}$$

(Overflow room for Problem 2, if needed)

3. (5 points) Suppose $T : P_1 \rightarrow P_2$ is a linear transformation with

$$T(2x+1) = 3x^2 - 1, \quad T(5x+3) = 2x+4.$$

Find $T(-1)$. Show all work.

$$\begin{aligned}
 -1 &= a(2x+1) + b(5x+3) \\
 2a+5b &= 0 \\
 -2(a+3b) &= -1 \\
 \hline
 0 - b &= 2 \Rightarrow b = -2 \\
 a &= 5
 \end{aligned}$$

check: $5(2x+1) - 2(5x+3) = -1$ ✓

$$T(-1) = T(5(2x+1) - 2(5x+3))$$

$$= 5T(2x+1) - 2T(5x+3)$$

$$= 5(3x^2 - 1) - 2(2x + 4) = \boxed{15x^2 - 4x - 13}$$

4. (6 points) Consider $\mathbb{R}^2$ and the following vector addition $\oplus$ and scalar multiplication $\odot$:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \oplus \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 - 2 \\ x_2 + y_2 + 3 \end{bmatrix}, \quad c \odot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} cx_1 - 2c + 2 \\ cx_2 + 3c - 3 \end{bmatrix}$$

a.) (1 point) Compute $\begin{bmatrix} 1 \\ -4 \end{bmatrix} \oplus \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1+0-2 \\ -4+2+3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$

b.) (3 points) Find the zero vector $\mathbf{0}$ for this vector addition $\oplus$. You must justify that $\mathbf{v} \oplus \mathbf{0} = \mathbf{v} = \mathbf{0} \oplus \mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^2$.

$$\mathbf{0} = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \text{ b/c}$$

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} \oplus \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2+x_1-2 \\ -3+x_2+3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \oplus \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} x_1+2-2 \\ x_2-3+3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \checkmark$$

c.) (2 points) Show the following vector space property holds:

$$1 \odot \mathbf{u} = \mathbf{u} \text{ for all } \mathbf{u} \text{ in } \mathbb{R}^2.$$

$$1 \odot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1x_1 - 2 \cdot 1 + 2 \\ 1x_2 + 3 \cdot 1 - 3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

5. (6 points) Consider the vector space M_{22} with standard matrix addition and scalar multiplication. Let O denote the 2×2 zero matrix. Consider the subset W of matrices A where $A^2 = O$, i.e.

$$W = \{A \text{ in } M_{22} \mid A^2 = O\}$$

Is W a subspace of M_{22} ? Fully justify your answer.

No. Consider: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ Both in W .

$$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \checkmark$$

But $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \neq O.$$

so $A+B$ is not in W .

so W is not a subspace of M_{22} .

6. (5 points) For the set $S = \left\{ \begin{bmatrix} c \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ c \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$, find all values of c for which the set S forms a basis for $\mathbb{R}^3$. Fully justify your answer.

Check lin. indep.

$$\begin{aligned}
 \det \begin{bmatrix} c & -1 & 1 \\ 0 & c & 2 \\ 1 & 1 & 2 \end{bmatrix} &= c \begin{vmatrix} c & 1 \\ 1 & 2 \end{vmatrix} - 2 \begin{vmatrix} c & -1 \\ 1 & 1 \end{vmatrix} \\
 &= c(2c-1) - 2(c+1) \\
 &= 2c^2 - c - 2c - 2 \\
 &= 2c^2 - 3c - 2 = 0 \\
 &(2c+1)(c-2) = 0 \\
 &c = -1/2, c = 2.
 \end{aligned}$$

If $c \neq -1/2, 2$, then $\det$ of matrix $\neq 0$, so vectors are lin. indep. 3 lin. indep vectors in $\mathbb{R}^3$ form a basis.

Alt. Soln:

$$\begin{bmatrix} c & -1 & 1 \\ 0 & c & 2 \\ 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & c & 2 \\ c & -1 & 1 \end{bmatrix} \xrightarrow{-cr_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & c & 2 \\ 0 & -c-1 & -2c+1 \end{bmatrix} \xrightarrow{r_2+r_3} \begin{bmatrix} 1 & 1 & 2 \\ 0 & c & 2 \\ 0 & -1 & -2c+3 \end{bmatrix}$$

$\swarrow \frac{1}{c}r_2 + r_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & c & 2 \\ 0 & 0 & -2c+3+\frac{2}{c} \end{bmatrix} \rightarrow 0$$

$$\begin{aligned}
 -2c + 3 + \frac{2}{c} &= 0 \\
 -2c^2 + 3c + 2 &= 0 \\
 c &= -1/2, 2.
 \end{aligned}$$

7. (8 points total) For this problem, use A and the reduced row echelon form of A , $RREF(A)$, to answer the following questions.

$$A = \begin{bmatrix} 1 & -1 & 1 & 2 & 0 \\ 2 & -4 & 0 & 3 & 2 \\ -3 & 4 & -2 & 1 & -1 \end{bmatrix} \quad RREF(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Consider the linear transformation $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$ (yes, this is the same A as above).

- a. (3 points) Find a basis for the kernel of T . Show all work.

$x_3 = s$
 $x_5 = t$
 $x_1 = -2s + t$
 $x_2 = -s + t$
 $x_4 = 0$

$$\begin{bmatrix} -2s+t \\ -s+t \\ s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Basis: $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

- b. (2 points) Is T one-to-one (also called *injective*)? Explain your answer in one sentence.

No, Ker has nontrivial elements (see basis).

- c. (1 point) Explain why $\left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \right\}$ forms a basis for the image of T .

These vectors are columns of A , and correspond to pivot columns in $RREF(A)$.

- d. (2 points) Is T onto (also called *surjective*)? Explain your answer in one sentence.

Yes! $\dim \text{im } T = 3 = \dim(\mathbb{R}^3)$

8. (6 points total) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples. Fully justify your answers.

a. (2 points) The set of vectors $W = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \mid ab \geq 0 \right\}$ is a subspace of $\mathbb{R}^2$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\downarrow$
 $1 \cdot 0 = 0$ ✓ $0 \cdot (-1) = 0$ ✓ $(1)(-1) < 0$ ✗

not a subspace b/c
not closed under add.

b. (2 points) Let V be $\mathbb{R}^2$ with the operation $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \oplus \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \end{bmatrix}$. $\oplus$ satisfies the vector space property $\mathbf{v} \oplus \mathbf{u} = \mathbf{u} \oplus \mathbf{v}$.

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \oplus \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-3 \\ 1-1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \oplus \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3-2 \\ 1-1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

not equal.

c. (2 points) The function $F : P_1 \rightarrow P_1$ by $F(ax + b) = ax + b^2$ is a linear transformation.

$$F(2(x+1)) = F(2x+2) = 2x+4$$

vs.

$$2(F(x+1)) = 2(x+1^2) = 2x+2$$

not equal.

Scratch paper page!!

Back of scratch paper page!